

ELEC 451 Power Electronics

Lab 1: Power Quality and Basic Uncontrolled Rectifiers

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October 1, 2026

I. INTRODUCTION

II. SIMULATION

- A. Circuit 1 — Full-Bridge Rectifier with Resistive Load
- B. Circuit 2 — Full-Bridge Rectifier with Capacitive Filter and Resistive Load
- C. Circuit 3 — Full-Bridge Rectifier with LC Filter and Resistive Load

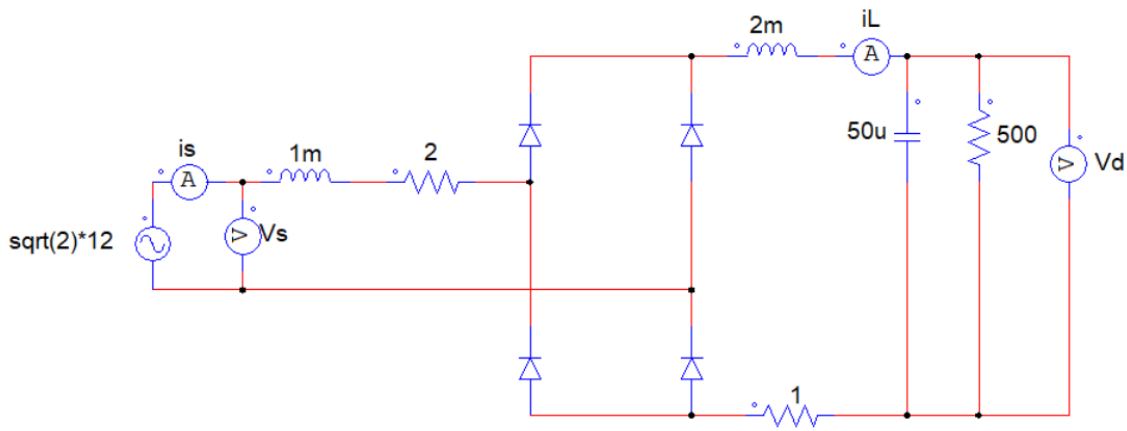


Fig. 1. PSIM schematic of Circuit 3.

The THD can be calculated as (these are the source voltage and current):

$$\text{THD} = \frac{\sqrt{\sum_{h \neq 1} I_{sh}^2}}{I_{s1}} \quad (1)$$

equivalently...

$$I_s = I_{s1} \sqrt{1 + \text{THD}^2} \quad (2)$$

$$\frac{I_s^2}{I_{s1}^2} - 1 = \text{THD}^2 \quad (3)$$

where I_{s1} is the fundamental frequency of the source current.

We know at steady state, the $I_s = I_L = I_o$. And since the load determines how much current flows through from the source,

$$I_{s1} = \frac{V_s}{R + R_{\text{shunt}}} \quad (4)$$

$$I_{s1} = \frac{12 \text{ V}}{500 + 1} \quad (5)$$

$$\boxed{I_{s1} = 23.95 \text{ mA}} \quad (6)$$

This is the fundamental component of $I_{s,\text{RMS}}$ since the harmonic distortion comes from the current ripple.

III. EXPERIMENT

- A. Circuit 1 — Resistive Load*
- B. Circuit 2 — Capacitive Filter and Resistive Load*
- C. Circuit 3 — LC Filter and Resistive Load*

IV. CONCLUSION